Experiment No-24 : Dieting System Based on Disease

Aim:

To write a Prolog program to suggest a dieting system based on the disease.

Programming Code:

```
% NOTE: For academic demonstration only. A real diet plan must be
%       prepared by a doctor or a qualified dietitian.

% ---------- Facts ----------
% recommend(Disease, ListOfFoodsToEat)
recommend(diabetes,     [whole_grains, green_vegetables, legumes, nuts, guava]).
recommend(hypertension, [fruits, vegetables, oats, banana, low_fat_milk]).
recommend(obesity,      [salads, vegetables, fruits, lean_protein, plenty_of_water]).
recommend(anemia,       [spinach, beetroot, dates, pomegranate, lentils, jaggery]).
recommend(heart_disease,[oats, fish, nuts, olive_oil, fruits, vegetables]).
recommend(constipation, [papaya, whole_grains, leafy_vegetables, curd, plenty_of_water]).

% avoid(Disease, ListOfFoodsToAvoid)
avoid(diabetes,     [sugar, sweets, soft_drinks, fried_foods]).
avoid(hypertension, [salt, pickles, processed_foods, fried_foods]).
avoid(obesity,      [fast_food, sugary_drinks, fried_foods, sweets]).
avoid(anemia,       [tea_with_meals, coffee_with_meals, junk_food]).
avoid(heart_disease,[butter, red_meat, fried_foods, excess_salt]).
avoid(constipation, [maida, junk_food, red_meat]).

% ---------- Rules ----------
should_eat(Disease, Food)   :- recommend(Disease, L), member(Food, L).
should_avoid(Disease, Food) :- avoid(Disease, L), member(Food, L).

% Diseases for which a given food must be avoided
avoid_for(Food, Disease) :- should_avoid(Disease, Food).

% Print the complete diet suggestion for a disease
suggest_diet(Disease) :-
    (   recommend(Disease, Eat), avoid(Disease, Avoid)
    ->  atomic_list_concat(Eat, ', ', EatText),
        atomic_list_concat(Avoid, ', ', AvoidText),
        format("Disease        : ~w~n", [Disease]),
        format("Foods to eat   : ~w~n", [EatText]),
        format("Foods to avoid : ~w~n", [AvoidText])
    ;   format("No diet information available for ~w.~n", [Disease])
    ).
```

OUTPUT:

```
$ swipl -q exp24.pl

?- suggest_diet(diabetes).
Disease        : diabetes
Foods to eat   : whole_grains, green_vegetables, legumes, nuts, guava
Foods to avoid : sugar, sweets, soft_drinks, fried_foods
true.

?- suggest_diet(anemia).
Disease        : anemia
Foods to eat   : spinach, beetroot, dates, pomegranate, lentils, jaggery
Foods to avoid : tea_with_meals, coffee_with_meals, junk_food
true.

?- suggest_diet(fever).
No diet information available for fever.
true.

?- should_eat(obesity, F).
F = salads ;
F = vegetables ;
F = fruits ;
F = lean_protein ;
F = plenty_of_water.

?- should_avoid(hypertension, F).
F = salt ;
F = pickles ;
F = processed_foods ;
F = fried_foods.

?- avoid_for(fried_foods, D).
D = diabetes ;
D = hypertension ;
D = obesity ;
D = heart_disease.

$
```

Result

The dieting system was implemented successfully using Prolog. Based on the disease entered, the program suggests the foods to be eaten and the foods to be avoided, and reports when no information is available for a disease.